

Plate Tectonics and Earthquakes

Plate tectonics explains how Earth's lithosphere is divided into moving plates that interact at boundaries.

Key facts:

Divergent boundaries form where plates move apart, often creating new oceanic crust at mid-ocean ridges.

Convergent boundaries form where plates move toward each other; subduction can produce trenches, volcanoes, and strong earthquakes.

Transform boundaries form where plates slide past each other, commonly generating shallow earthquakes.

Most earthquakes occur when stress on a fault exceeds friction and stored elastic energy is released suddenly.

The focus is the point inside Earth where rupture begins, and the epicenter is the surface point above the focus.

P waves travel fastest and pass through solids, liquids, and gases; S waves travel only through solids.

Seismic waves and shadow zones help scientists infer the structure of Earth's interior.

Answerable questions:

Q: What causes many earthquakes? A: Many earthquakes are caused by sudden slip on faults at or near tectonic plate boundaries.

Q: Why are S waves useful? A: Because S waves do not travel through liquids, their paths provide evidence about liquid layers inside Earth.